26/10/2026
Monday

Activity

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1) A CCTV Image Contains salt and pepper noise due to transmission errors

Technique: Median filter

Reason:- Removes Black and White noise by replacing each pixel with the median of its neighbouring pixels while preserving edges

2) A Satellite Image is corrupted by Gaussian noise from sensors.

Technique: Gaussian filtering

Gaussian

Reason:- Reduces Noise by Smoothing random intensity variations in the Image.

3) A Medical X-ray Image has random noise, but fine edges must be preserved.

Technique: Bilateral filtering

Reason:- Remove noise while preserving edges by considering spatial distance and pixel intensity differences.

4) A scanned document has BLACK and White dots scattered randomly.

technique: Median filtering

Reason: Removes isolated BLACK and White dots while preserving text boundaries

5) A low-light image shows heavy grain without clear Boundaries

Technique: Non-local Means Denoising

Reason:- Reduces image grain by averaging similar image patches while preserving important details.

6) A foggy road image has very low Contrast

Technique: CLAHE (Contrast limited Adaptive Histogram Equalization)

Reason:-

Enhances local Contrast and improves the visibility of road objects.

7) A underwater image appears dull and washed out

Technique: White Balance and Contrast Enhancement

Reason: Correct Color Imbalance and improve image Contrast and visibility.

8) A Chest x-ray Slow poor visibility of Internal Structure

Technique: Histogram Equalization

Reason:- Redistributes pixel Intensity values to

improve Contrast and reveal Internal details

9) A Night time traffic image Lacks Intensity Variation.

Technique: Gamma Correction

Reason: Adjusts brightness nonlinearly to make dark objects more visible.

10) A grayscale image uses only a Narrow Intensity Image

Technique: Contrast Stretching

Reason: Expands the used Intensity range to improve the overall image Contrast.